

Influence of dietary factors on the clinical course of ulcerative colitis: a prospective cohort study

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Background and aims: The causes of relapses of ulcerative colitis (UC) are unknown. Dietary factors have been implicated in the pathogenesis of UC. The aim of this study was to determine which dietary factors are associated with an increased risk of relapse of UC. Methods: A prospective cohort study was performed with UC patients in remission, recruited from two district general hospitals, who were followed for one year to determine the effect of habitual diet on relapse. Relapse was defined using a validated disease activity index. Nutrient intake was assessed using a food frequency questionnaire and categorised into tertiles. Adjusted odds ratios for relapse were determined using multivariate logistic regression, controlling for non-dietary factors. Results: A total of 191 patients were recruited and 96% completed the study. Fifty two per cent of patients relapsed. Consumption of meat (odds ratio (OR) 3.2 (95% confidence intervals (CI) 1.3–7.8)), particularly red and processed meat (OR 5.19 (95% CI 2.1–12.9)), protein (OR 3.00 (95% CI 1.25–7.19)), and alcohol (OR 2.71 (95% CI 1.1–6.67)) in the top tertile of intake increased the likelihood of relapse compared with the bottom tertile of intake. High sulphur (OR 2.76 (95% CI 1.19–6.4)) or sulphate (OR 2.6 (95% CI 1.08–6.3)) intakes were also associated with relapse and may offer an explanation for the observed increased likelihood of relapse. Conclusions: Potentially modifiable dietary factors, such as a high meat or alcoholic beverage intake, have been identified that are associated with an increased likelihood of relapse for UC patients. Further studies are needed to determine if it is the sulphur compounds within these foods that mediates the likelihood of relapse and if reducing their intake would reduce relapse frequency.